Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/22/2022
PI Name:	St Hilaire, Melissa Ph.D.		
Project Title:	Towards a Real-Time Estimate of Circadian Phase During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	02/01/2020	**End Date:**	01/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2022 per NSSC information (Ed., 7/9/21) NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 2/3/21)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Lockley, Steven Ph.D. (Brigham And Women's Hospital, Inc.) Rahman, Shadab Ph.D. (Brigham And Women's Hospital, Inc.)		
Grant/Contract No.:	80NSSC20K0576		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The 24-hour circadian clock, located in the hypothalamus of the brain, controls the timing and pattern of sleep, alertness, performance, and mood, disruption of which has a major impact on safety. The circadian clock also controls many other aspects of human physiology, perturbation of which will affect longer-term health and well-being. It is imperative that the timing of the circadian clock is measured routinely, as a standard measure of health and behavior, during long-duration space missions to ensure that these multiple brain and body systems are not disrupted and misaligned from either each other (internal desynchronization) or with the work schedule (external desynchronization). Once measured, and if out of phase, circadian rhythms can be resynchronized to the desired time using a number of countermeasures, including bright light, melatonin, and potentially exercise and meal timing. The effectiveness of these countermeasures depends critically, however, on the circadian phase at which they are administered; mistimed countermeasures can exacerbate circadian disruption, leading to worsening of health and performance outcomes rather than improvement. Circadian phase is highly individualized, however, varying by up to 5 hours in healthy individuals under stable conditions, and is even more variable in space. To date, no feasible method exists to measure circadian phase accurately and reliably in real-time during spaceflight. Existing methods currently being used in-flight require collection of serial urine samples for up to 48 hours, from which the concentration of the urinary metabolite 6-sulphatoxymelatonin is measured to determine the timing of the circadian rhythm. Importantly, however, no technology exists to measure concentrations of urinary 6-sulphatoxymelatonin directly in-flight; therefore, knowledge about in-flight circadian disruption can be determined only during post-hoc analyses once the samples have been returned to Earth, which limits the ability to treat circadian disruption in real time. This proposal addresses this critical gap by (1) validating the use of commercially available meters to derive circadian phase in real-time from other biomarkers that exhibit robust circadian rhythms; and (2) developing a metric that estimates circadian phase from only two samples, thereby minimizing the need for serial data collection. The successful validation of this approach will enable circadian phase to be measured in real-time during spaceflight with minimal burden to the crew such that countermeasures to mitigate the effects of circadian misalignment can be tailored to individual crewmembers needs, thereby enhancing the safety, performance, and well-being of all crewmembers during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA has invested significant research into the design of countermeasures to mitigate the effects of circadian misalignment. These countermeasures will be ineffective if not timed appropriately relative to circadian phase. Our real-time circadian phase estimate is expected to significantly improve the impact of such countermeasures. In addition, treatments for a wide range of conditions, including circadian rhythm sleep disorders, hypertension, and cancer, are maximally effective when timed relative to circadian phase, yet circadian phase remains absent from standard clinical assessments due to significant time and cost constraints in obtaining an accurate measure. The ability to estimate circadian phase non-invasively in real-time over a short collection window is therefore expected to enable ‘circadian medicine’ within standard clinical practice.
Task Progress:	Aim 1: Validate the use of real-time measures of saliva cortisol and blood cholesterol to derive circadian phase. We have enrolled 19 (10 males, 9 females) of the targeted 20 participants into this study, and have completed sample collections in 13 participants. Participants are collecting several sample types (blood, urine, saliva) approximately every 3 hours over two 48-hour periods at home. We are collecting samples to assess several biomarkers using real-time assays via commercially available point-of-care devices as well as traditional assays for validation. Recruitment, data collection, and data analyses are ongoing. Aim 2: Develop a method to derive circadian phase from as few as three sample time points. Due to delays in sample collection due to COVID-19, we have started algorithm development and validation on available inpatient laboratory data. We have developed our preliminary approach to estimate circadian phase from biomarker concentrations measured from three samples collected over a 16-hour period. We have compared circadian phase estimates from this approach to the gold-standard marker of circadian phase, the dim light melatonin onset from plasma melatonin, and can estimate a significant proportion of circadian phases to within 45 minutes of the gold-standard marker. We will further refine, validate, and test this method with the real-time data collected in the current study as results are assayed and available.
Bibliography Type:	Description: (Last Updated: 12/06/2023)
Articles in Peer-reviewed Journals	St Hilaire MA, Lammers-van der Holst HM, Chinoy ED, Isherwood CM, Duffy JF. "Prediction of individual differences in circadian adaptation to night work among older adults: application of a mathematical model using individual sleep-wake and light exposure data." Chronobiol Int. 2020;37(9-10):1404-11. https://doi.org/10.1080/07420528.2020.1813153 ; PMID: 32893681 , Sep-2020
Articles in Peer-reviewed Journals	Klerman EB, Rahman SA, St Hilaire MA. "What time is it? A tale of three clocks, with implications for personalized medicine." J Pineal Res. 2020 May;68(4):e12646. https://doi.org/10.1111/jpi.12646 ; PMID: 32155668; PMCID: PMC7285860 , May-2020